4-Zone Conventional Fire Alarm Panel — FINAL Architecture & Power Tree (V05)

Scope: 4-zone conventional panel, 25-30 field devices, single-MCU architecture (no STM8, no shift registers), RS-485 present, talkback/GSM/multi-footprint OUT of scope for this prototype.
Basis: Ravel RE-2554/2558 reverse-engineering (Rev-Engg-Power-Circuit.xlsx), EMKIONYX BOM v1, existing hardware/electrical docs (mcu_selection.md, power_budget.md, system block diagrams).

SHEETS IN THIS WORKBOOK

Architecture	- block-level design, single-MCU GPIO budget, what changed vs the donor design
Quantity_BOM	- component quantities sized for 4 zones / 25-30 devices, single-MCU version
Power_Tree	- hierarchical Source -> Regulator -> Rail -> Load current/power model
Open_Decisions	- the 2 items that block a final power budget until you confirm them

HOW TO USE

YELLOW cells in Power_Tree (per-load current assumptions) and Open_Decisions.
Every other cell is a formula that rolls child-load current up to its parent rail automatically.
Two scenarios are computed side by side: STANDBY and ALARM (worst case, per EN54-4 style budgeting)

KEY CHANGES FROM THE REVERSE-ENGINEERED (Ravel-derived) DESIGN

1. STM8S001J3 (display MCU) removed - STM32G0B1CBT6 alone drives ADC, RS-485, relays, NAC, and HMI.
2. 8x HEF4094BT shift registers removed - 12 LEDs + 9 keys + I2C LCD fit directly on STM32G0B1 GPIO.
3. Donor's main MCU; STM32G0B1 replaces it too.
4. L7824CV (30V->24V) removed - your SMPS already outputs 24V, so there is nothing to regulate down.
   This assumes your SMPS output IS the 24V board rail.
5. Power_Tree is built assuming the 3.3V part.

WHAT THIS WORKBOOK DOES NOT YET DO

- It does not confirm battery charge voltage headroom - see Open_Decisions, item 1.
- LCD current is an assumption (generic I2C character LCD, backlight on).

System Architecture — 4-Zone Conventional Panel, Single-MCU

1. SIGNAL FLOW (confirmed topology)

AC Mains -> SMPS module (bought-out, external) -> 30V DC raw output
30V raw -> L7824CV (linear reg) -> 24V regulated board rail (zones, relay coils, NAC, pre-reg chain)
30V raw -> LM317T (CC/CV charger, set 27.6-28.2V float) -> 2x 12V 7Ah SLA in series (24V battery string)
24V rail -> L7812CV (pre-regulator / thermal buffer) -> 12V rail (buzzer oscillator, feeds 5V stage)
12V rail -> L7805CV -> 5V rail (driver array logic, external LCD, MAX485 + level shifter)
5V rail -> 3.3V LDO -> 3.3V rail (STM32G0B1CBT6, indicator LEDs, keypad, level-shifter 3.3V side)
STM32G0B1 <-> level shifter <-> MAX485ESA+ <-> RS-485 A/B terminal block (screw terminals, field bus)
STM32G0B1 GPIO -> TBD62083 (sink driver) -> 3x relay coils (Fire/Fault/Supervisory) on 24V
STM32G0B1 GPIO -> TIP127 PNP Darlington x2 (high-side) -> NAC1/NAC2 on 24V, EOL-supervised
STM32G0B1 ADC (8ch) <- voltage dividers <- Zone 1-4 loops, battery sense, mains/charger sense, rail health
STM32G0B1 GPIO -> Q_reset (NPN) -> shared zone-reset line -> drops 24V briefly to unlatch detectors
STM32G0B1 I2C -> external LCD module (bought-out)
STM32G0B1 GPIO (direct, no shift register) -> 12 indicator LEDs + 9 keys

2. WHY THIS DIFFERS FROM THE DONOR (RAVEL RE-2554/2558) BOM

Donor functional details	Removed /	Reason
Obsolete MCU part	STM32G0B1CBT6 (already your	12-bit ADC vs 10-bit; dual watchdog + HW CRC for EN54-2 cl.13; active lifecycle
Redundant part	Removed — STM32G0B1 drives	Only 1 MCU needed at 4-zone scale; removes inter-MCU UART link and second firmware image
Shift registers	Removed — direct GPIO	12 LEDs + 9 keys fit STM32G0B1's ~37 usable GPIO with margin;
30V->24V	KEPT	30V raw SMPS output gives the LM317T charger headroom to hit 27.6-28.2V float; L7824CV makes the clean 24V board rail
5V RS-485 transceiver	KEPT + level shifter added	level shifter (3.3V<->5V) bridges STM32G0B1's 3.3V UART to the 5V transceiver
Darlington Arrays Sink Drivers	Reduced to x1	Only 3 relay coils + 2 NAC-enable + buzzer-gate = 6 channels; one 8-channel driver covers it without the LED-matrix channels the donor needed

3. STM32G0B1CBT6 GPIO BUDGET (LQFP48)

Function	Pins used	Notes

Zone ADC inputs (4 zones)	4	ADC1_IN, 12-bit, via divider network
Battery / mains / charger / rail-health sense	4	ADC1_IN, shared ADC peripheral
RS-485 UART + DE/RE	3	USART TX, RX, DE/RE via level shifter
Relay drive (Fire/Fault/Supervisory, via TBD62083)	3	GPIO out, sinks through driver
NAC1/NAC2 drive (via TIP127)	2	GPIO out
Zone reset	1	GPIO out -> Q_reset base
Buzzer gate (silence control)	1	GPIO out, gates NE555 output or reset
Indicator LEDs	12	Direct GPIO, current-limit resistor per LED
Keys (Silence, Menu, Enter, Evacuate, Lamp Test, cursor L/R, 2x spare/alphanumeric)	9	GPIO in, pull-up, debounce in firmware
LCD interface — JHD162A (parallel HD44780, 4-bit mode)	6	RS, E, D4-D7. NOTE: JHD162A is a native parallel-interface LCD, not I2C — 6 GPIO in 4-bit mode, not 2. Add a PCF8574 I2C backpack module if you want to free up 4 GPIO instead.
SWD (program/debug)	2	SWDIO, SWCLK — reserved, not app GPIO
Crystal/oscillator (if used)	2	OSC_IN/OUT — reserved if external crystal used
TOTAL used	49	of ~48 physical pins on LQFP48 — JHD162A's parallel interface pushes this over. Either add a PCF8574 I2C backpack to the LCD (frees 4 pins, back to comfortable margin), or drop 4 of the 12 LEDs / share the 9 keys as a row-column matrix (4 pins instead of 9). Flagged in Open_Decisions.

Note:

Relay coil drivers are Q1/Q2 = BDX34C (PNP Darlington), Relay part is G5V-2

4. VOLTAGE MARGIN ANALYSIS (regulator output vs downstream device spec)

Rail / Node	Regulator (confirmed part)	Output spec	Downstream device	Margin	Source / Note

30V raw -> 24V rail	L7824CV, linear reg	Fixed 24V, typ. tolerance per L78xx family (+/-4% class)	Field devices, relay coils (G5V-2, 24V coil), NAC EOL circuits all rated 24V nominal	Comfortable — 24V devices rated for 20-28V typ. range	L7824CV output tolerance figure is generic L78xx family spec; confirm on manufacturer datasheet at BOM lock (Open_Decisions #13)
24V -> 12V rail	L7812CV, linear reg (pre-regulator/thermal buffer)	Fixed 12V, typ. +/-4% class	NE555 buzzer oscillator: rated 4.5-16V supply (generic NE555 spec)	Wide margin both sides	Input-output differential (24V->12V) dissipates real power in L7812CV at Alarm current — confirm thermal design at bring-up
12V -> 5V rail	L7805CV, linear reg	Fixed 5V, typ. +/-4% class (4.8-5.2V)	MAX485ESA+ Vcc spec: 4.75-5.25V (industry-standard RS-485 transceiver spec); JHD162A LCD: 5V logic; TBD62083AFWG logic Vcc: 4.5-5.5V	Tight but positive on both bounds (4.8V rail min vs 4.75V device min)	Confirm L7805CV grade/tolerance on manufacturer datasheet at BOM lock — margin is the tightest in the chain (Open_Decisions #13)
5V -> 3.3V rail	AMS1117-3.3 (SOT-223), confirmed part	Vout 3.300V typ; 3.251-3.349V (IOUT=0.8A window) or 3.201-3.399V (full spec window) per AMS1117 datasheet Electrical Characteristics table	STM32G0B1CBT6 VDD/VDDA: 1.7-3.6 V; VDDIO2: 1.65-3.6V (datasheet Section "Reset and power management")	Very wide margin — 3.3V sits well inside the 1.7-3.6 V window on every corner	AMS1117 datasheet, Electrical Characteristics, Output Voltage row
5V -> 3.3V rail, dropout check	AMS1117-3.3	Dropout voltage: 1.1V typ / 1.3V max at IOUT=0.8A (Note 4)	Actual load at this rail (3.3V rail total) is 36mA standby / 91mA alarm per Power_Tree — far below the 0.8A test point	Vin-Vout available = 5.0-3.3 = 1.7V, exceeds the 1.3V max-current dropout spec with 0.4V to spare, and dropout is lower still at the 36-91mA actual load	AMS1117 datasheet, Electrical Characteristics, Dropout Voltage row + Application Hints
AMS1117-3.3 current limit	AMS1117-3.3	Current limit: 900 mA min / 1500 mA max (Vin-Vout=1.5V)	3.3V rail total draw: 36mA standby / 91mA alarm	Over 10x margin below the current limit — no risk of the LDO folding back under this load	AMS1117 datasheet, Electrical Characteristics, Current Limit row

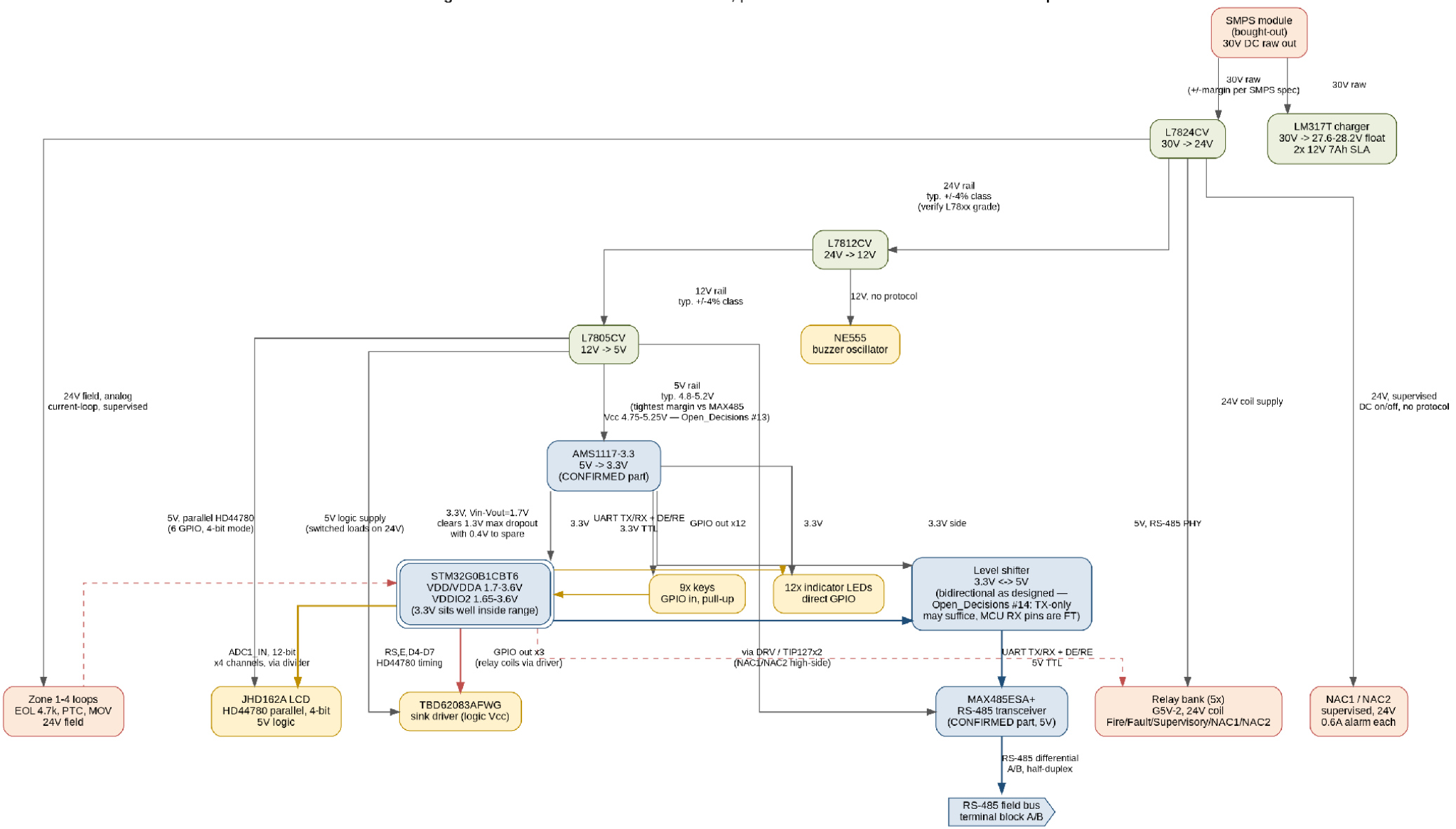
5. COMMUNICATION / SIGNAL PROTOCOLS (by interface)

Interface	Standard / Physical layer	Endpoints	Voltage domain(s)	Typical rate	Note
Field bus (external, panel-to-panel or panel-to-repeater)	RS-485, differential, 2-wire (A/B)	MAX485ESA+ <-> external RS-485 bus terminal block	5V logic side (MAX485), 24V-referenced field wiring not applicable — RS-485 is its own differential pair, isolated from the 24V rail	Application-defined (typ. 9600-115200 bps for panel telemetry)	MAX485ESA+ is a 5V part; half-duplex, DE/RE tied together at the transceiver, driven by one MCU GPIO
MCU UART <-> RS-485 transceiver	UART, TTL levels, TX/RX + DE/RE	STM32G0B1CBT6 USART <-> level shifter <-> MAX485ESA+	3.3V (MCU side) <-> 5V (MAX485 side), bridged by the level shifter (U3)	Matches the RS-485 bus rate above	STM32G0B1 UART pins are FT (5V-tolerant) per datasheet Table 12 — RX into the MCU could take 5V directly; TX out of the MCU (3.3V high) is the side that actually needs level-shifting up to guarantee MAX485 VIH — see Open_Decisions #14 for a possible simplification
Zone loops (4x)	Analog current-loop, supervised (EOL 4.7 k)	Zone 1-4 field wiring <-> voltage divider <-> STM32G0B1 ADC1_IN	24V field side, divided down into the 0-3.3V ADC input window	Polled — no fixed bit rate, ADC sample rate per firmware scan loop	12-bit ADC1_IN per zone; divider network sized so full-scale zone loop voltage maps below 3.3V VDDA
HMI display	HD44780 parallel, 4-bit mode	STM32G0B1 GPIO (RS, E, D4-D7) <-> JHD162A LCD	3.3V MCU side driving a 5V-logic LCD module — JHD162A accepts TTL-level inputs at VIH>=2.2V typ, compatible with 3.3 V GPIO high (~3.0V min)	Parallel, no clocked bit rate — E-strobe timed in firmware per HD44780 timing spec	NOT I2C as earlier assumed — native parallel HD44780 interface, 6 GPIO in 4-bit mode (see Open_Decisions #6)

Relay / NAC drive	GPIO -> driver IC -> switched load (no data protocol)	STM32G0B1 GPIO <-> TBD62083AFWG (sink) / TIP127 (high-side) <-> relay coils / NAC lines	3.3V GPIO logic controlling 24V switched loads through the driver stage	N/A — DC on/off control, not a data bus	TBD62083AFWG logic supply is 5V (per BOM); GPIO drive is 3.3V-compatible at its logic input per typical sink-driver input spec — confirm exact part's VIH against 3.3V GPIO at BOM lock
SWD (program/debug)	Serial Wire Debug, 2-wire (SWDIO/SWCLK)	STM32G0B1 <-> external programmer/debugger header	3.3V logic	Debug-only, not a runtime interface	Reserved pins, not part of the application GPIO budget

Architecture Block Diagram (V05) — rail voltages, tolerances, and protocols per interface

Source rendered from block_diagram.dot; also delivered standalone as FAP_Architecture_Block_Diagram.png / .svg for direct reference when drawing the schematic in KiCad.



Quantity BOM — sized for 4 zones / 25-30 field devices, single-MCU

Component	Package/Type	Qty	Rail	Ref Des	Sizing basis
STM32G0B1CBT6	LQFP48	1	3.3V	U1	Single MCU — see Architecture sheet
MAX485ESA+	SOIC-8	1	5V	U2	RS-485 transceiver, kept per your confirmation
Level shifter (3.3V<->5V, e.g. TXB0104 or BSS138 4-ch module)	TSSOP14/module	1	3.3V/5V	U3	Bridges MCU UART+DE/RE to MAX485ESA+ (part CONFIRMED, 5V RS-485 transceiver, kept per your decision — see Open_Decisions #2 and #14)
L7824CV	TO-220	1	30V->24V	U4	30V raw -> 24V board rail
LM317T	TO-220	1	30V->28V	U5	CC/CV charger, 27.6-28.2V float, 0.8A max
L7812CV	TO-220	1	24V->12V	U6	Pre-regulator / thermal buffer
L7805CV	TO-220	1	12V->5V	U7	5V logic rail
AMS1117-3.3	SOT-223	1	5V->3.3V	U8	MCU + LED + keypad rail. Part CONFIRMED
TBD62083AFWG (8-ch sink driver)	SOP-18	1	5V logic / 24V switched	U9	3 relay coils + 2 NAC-enable + buzzer gate = 6 of 8 channels used
G5V-2 relay, 24V coil (Form C)	TO5/PCB relay	5	24V coil (confirmed)	K1-K5	Fire, Fault, Supervisory + NAC1, NAC2. Part number G5V-2 per actual .kicad_sch. Coil voltage CONFIRMED as 24V (your decision) — order the 24VDC-coil variant of G5V-2, not 5V/12V
Flyback diode (1N4148/1N4007)	DO-35/DO-41	5	24V	D1-D5	One per relay coil
TIP127 (PNP Darlington, relay coil driver)	TO-220	2	24V coil drive	Q1, Q2	CONFIRMED by direct visual inspection of the physical Ravel board. The other engineer's .kicad_sch showed 'BDX34C' at this position — checked both datasheets: same TO-220 package, same Base-Collector-Emitter pinout, same 100V rating, both wildly overspecified for a relay coil's actual current, so no functional difference either way. Going with what you saw on the real hardware.

NE555	SOIC-8	1	12V	U10	MCU-independent fault buzzer oscillator
Piezo buzzer element	THT	1	12V (via	BZ1	Driven by NE555
NPN transistor (e.g. MMBT2222A)	SOT-23	1	24V switched	Q3	Shared zone-reset drive line
PPTC resettable fuse	THT/SMD	8	24V	F1-F8	1 per zone (4) + 1 per NAC (2) + battery input (1) + RS-485 bus (1)
MOV (varistor, 24V-rated)	THT/SMD	10	24V/field	MOV1-10	2 per zone (4x2=8) + 1 per NAC (2) — per donor topology
EOL resistor 4.7k ohm 1/4W	THT	6	Field	R_EOL1-6	1 per zone (4) + 1 per NAC (2); matches Ravel RE-RP spec (4.7K, 1/4W)
Zone divider resistor network (2x per zone)	THT/SMD	8	24V->ADC	R1-R8	Scales zone loop voltage below 3.3V ADC full-scale
LED, 3mm/5mm, various colours	THT	12	3.3V	LD1-LD12	4 zone Fire LEDs + System ON/AC ON/Battery ON/Charger ON/Fire/Fault/Supervisory/Silenced
Tactile switch	THT/SMD	9	3.3V	SW1-SW9	Silence, Menu, Enter, Evacuate, Lamp Test, cursor L/R, 2x spare/alphanumeric
External LCD module (bought-out)	Module	1	5V (assumed)	LCD1	I2C character LCD; confirm exact part + current draw
2-pin terminal block, 5.08mm pitch	THT	9	Field/24V	J1-J9	Zone1-4 (4) + NAC1-2 (2) + Battery in (1) + SMPS/30V in (1) + spare (1)
2-pin terminal block (RS-485 A/B)	THT	1	Field	J10	RS-485 bus
2-pin terminal block (RS-SEN)	THT	1	24V/sense	J12	repeater/remote sense input, divider-fed to analog mux
2-pin terminal block (RS-24VOP)	THT	1	24V	J13	CONFIRMED from actual .kicad_sch — direct 24V output, likely for repeater panel power
3-pin terminal block (C/NO/NC)	THT	3	24V switched	J11-J13	Fire, Fault, Supervisory relay contacts
TVS diode (RS-485 bus protection)	SMD	2	Field	D4, D5	A/B line transient protection
Decoupling/bulk capacitors (assorted)	SMD/THT	~35	All rails	C-misc	100nF per IC pin + bulk 10-470uF per regulator stage — itemize at schematic capture
2x 12V 7Ah SLA battery	External	2	24V string	BAT1, BAT2	24h standby + 30min alarm sizing

Power Tree — Source -> Regulator -> Rail -> Load

Yellow cells = your editable assumptions. Everything else is a formula. Two scenarios: STANDBY and ALARM (worst case).

	Node / Load	Voltage (V)	I Standby (mA)	I Alarm (mA)	P Standby (mW)	P Alarm (mW)	Notes
3.3V RAIL (from AMS1117-3.3, confirmed part)							
1	STM32G0B1CBT6 (Run mode, ADC+USART+GPIO active)	3.3	10	16	33	52.8	Datasheet-verified Table 28 base (~2-9mA depending on fHCLK/Range) + peripheral adders (Table 36) for ADC/USART/GPIO; alarm = higher activity, faster ADC
2	Indicator LEDs (12x, direct GPIO, current-limit R)	3.3	24	72	79.2	237.6	Standby: 4 system LEDs lit (System/AC/Battery/Charger ON) @ ~6mA. Alarm: all 12 lit worst case @ ~6mA
3	Keypad pull-ups/pull-downs (9 keys)	3.3	1	1	3.3	3.3	Negligible, internal MCU pull-ups
4	Level shifter, 3.3V side	3.3	1	2	3.3	6.6	Logic-side draw only; 5V side counted on 5V rail
	3.3V RAIL TOTAL (LDO output)	**3.3**	**36**	**91**	**118.8**	**300.3**	*Sum of all 3.3V loads above*
5V RAIL (from L7805CV)							
1	TBD62083AFWG driver, logic supply only	5	5	5	25	25	Logic Icc only — switched relay/NAC current flows on 24V, not through this rail
2	External LCD module — JHD162A (confirmed)	5	1.5	1.5	7.5	7.5	From JHD162A datasheet: +5V, 1.5 mA. Replaces earlier generic-LCD assumption.
3	MAX485ESA+ RS-485 transceiver	5	2	4	10	20	Receiver-enabled idle ~0.3-1mA per datasheet; alarm scenario includes occasional TX bursts, engineering margin applied
4	Level shifter, 5V side	5	1	2	5	10	
5	3.3V LDO input draw (= 3.3V rail current, reflected)	5	36	91	180	455	Linear LDO: Iin approx Iout (dropout small at these currents); formula-linked below

Power Tree — Source -> Regulator -> Rail -> Load

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	Node / Load	Voltage (V)	I Standby (mA)	I Alarm (mA)	P Standby (mW)	P Alarm (mW)	Notes
	5V RAIL TOTAL (L7805CV output)	**5**	**45.5**	**103.5**	**227.5**	**517.5**	*Sum of all 5V loads above, including reflected 3.3V rail current through the LDO*
	12V RAIL (from L7812CV — pre-regulator / thermal buffer)						
1	NE555 buzzer oscillator	12	3	15	36	180	Standby: quiescent bias only (silent). Alarm/fault: actively sounding, driving piezo
2	L7805CV input draw (= 5V rail current, reflected)	12	45.5	103.5	546	1242	
0	**12V RAIL TOTAL (L7812CV output)**	**12**	**48.5**	**118.5**	**582**	**1422**	*Sum of 12V loads, including reflected 5V rail current through L7805CV*
	24V RAIL (from L7824CV — 30V raw -> 24V board rail)						
	Field device / zone-loop assumptions (feeds the 4 zone rows below via formula)						
1	EOL resistor value (ohm)		4700				
2	Devices per zone (avg, 25-30 total / 4 zones)		7				
3	Device standby current (mA each) — 2-wire 24VDC detector spec		0.1				
4	Device alarm current (mA, one device activated — loop is current-limited)		25				

Power Tree — Source -> Regulator -> Rail -> Load

Yellow cells = your editable assumptions. Everything else is a formula. Two scenarios: STANDBY and ALARM (worst case).

	Node / Load	Voltage (V)	I Standby (mA)	I Alarm (mA)	P Standby (mW)	P Alarm (mW)	Notes
5	Zone 1 loop (EOL + real detector spec)	24	5.806382979	30.10638298	139.3531915	722.5531915	Standby = EOL current + (devices x 0.1mA). Alarm = EOL current + 25mA (worst-case single device; the loop's 500-ohm current-limit divider caps combined multi-device alarm current well below a simple N x 25mA sum)
6	Zone 2 loop	24	5.806382979	30.10638298	139.3531915	722.5531915	Same formula as Zone 1
7	Zone 3 loop	24	5.806382979	30.10638298	139.3531915	722.5531915	Same formula as Zone 1
8	Zone 4 loop	24	5.806382979	30.10638298	139.3531915	722.5531915	Same formula as Zone 1
9	NAC1 (supervised, EOL 4.7k)	24	5	600	120	14400	Standby = supervision current only; alarm = full 0.6A sounder load
10	NAC2 (supervised, EOL 4.7k)	24	5	600	120	14400	Same basis as NAC1
11	Relay coil — RL3 Fire (de-energised in normal)	24	0	15	0	360	24V coil CONFIRMED (your decision). ~360mW coil, energises only in alarm
12	Relay coil — RL4 Fault (fail-safe: energised in normal)	24	15	15	360	360	24V coil CONFIRMED. Energised-in-normal is deliberate (dead panel must report itself as dead) — do not remove
13	Relay coil — RL5 Supervisory (energised in normal)	24	15	15	360	360	24V coil CONFIRMED. Also gates the buzzer per netlist (BUZ-1/BUZ-2 wired through RL5 contacts)
14	Relay coil — RL1 NAC1 (de-energised in normal)	24	0	15	0	360	24V coil CONFIRMED. Driven by Q1 TIP127 (confirmed by direct inspection of the physical board — see Quantity_BOM note)

Power Tree — Source -> Regulator -> Rail -> Load

Yellow cells = your editable assumptions. Everything else is a formula. Two scenarios: STANDBY and ALARM (worst case).

	Node / Load	Voltage (V)	I Standby (mA)	I Alarm (mA)	P Standby (mW)	P Alarm (mW)	Notes
15	Relay coil — RL2 NAC2 (de-energised in normal)	24	0	15	0	360	24V coil CONFIRMED. Driven by Q2 TIP127 (same confirmation as RL1)
16	Zone reset transistor drive	24	0	0	0	0	Pulsed only during reset command, negligible average current
17	RS-SEN sense divider (J12, R69 820k + R70 4.7k)	24	0.03	0.03	0.72	0.72	2-pin terminal, divider feeds analog mux (U12/U13) — supervises repeater link presence/health. Bias current only, negligible
18	RS-24VOP output (J13, direct 24V tap)	24	0	0	0	0	2-pin terminal, direct 24V output — no relay switching. 0mA in this prototype (no repeater connected yet). If a repeater is wired in later, add ~50-70mA per RE-RP datasheet
19	L7812CV input draw (= 12V rail current, reflected)	24	48.5	118.5	1164	2844	
	24V RAIL TOTAL (L7824CV output)	**24**	**111.7555319**	**1513.955532**	**2682.132766**	**36334.93277**	*Sum of all 24V loads, including reflected 12V/5V/3.3V chain current through L7812CV*
20	L7824CV own quiescent current (own overhead, on 30V side)	30	5	5	150	150	Typical linear regulator IQ, drawn from the 30V input side

30V RAW (SMPS output, before L7824CV / LM317T)							
1	L7824CV input draw (= 24V rail current, reflected)	30	111.7555319	1513.955532	3352.665957	45418.66596	
2	L7824CV own quiescent current	30	5	5	150	150	
3	LM317T charger — quiescent (adjust pin + sense dividers)	30	8	8	240	240	Present continuously regardless of charge state

Power Tree — Source -> Regulator -> Rail -> Load

Yellow cells = your editable assumptions. Everything else is a formula. Two scenarios: STANDBY and ALARM (worst case).

	Node / Load	Voltage (V)	I Standby (mA)	I Alarm (mA)	P Standby (mW)	P Alarm (mW)	Notes
4	LM317T charger — active charge current	30	0	800	0	24000	0 in normal standby (battery topped up); up to 0.8A max during a recharge cycle, which is the PSU worst case, not the alarm case, per power_budget.md methodology
	30V RAW TOTAL DRAW (from SMPS)	**30**	**124.7555319**	**2326.955532**	**3742.665957**	**69808.66596**	*This is the current the SMPS module must actually deliver*

SMPS SIZING — YOUR OWN CALCULATION (not copied from Ravel's rating)

	Design margin over worst-case draw		0.25	*(edit — 25% shown, matches the derating style)*
	Worst case = ALARM + active battery recharge simultaneously		**124.7555319**	**2326.955532**
	Recommended SMPS continuous rating (mA), with margin			**2908.694415**
	Recommended SMPS rating (A), with margin			**2.91**

Buy an SMPS module rated 30V DC output at or above the computed Amps figure above. This is the req number, derived bottom-up from this panel's actual loads

BATTERY SIZING (24V string)

	Standby current on 24V-equivalent bus (mA)		111.7555319
	Alarm current on 24V-equivalent bus (mA)		1513.955532

Power Tree — Source -> Regulator -> Rail -> Load

Yellow cells = your editable assumptions. Everything else is a formula. Two scenarios: STANDBY and ALARM (worst case).

	Node / Load	Voltage (V)	I Standby (mA)	I Alarm (mA)	P Standby (mW)	P Alarm (mW)	Notes
	Standby duration required (h)		24				
	Alarm duration required (h)		0.5				
	Derating factor (ageing/temp/EOD)		1.25				
	Required battery capacity (Ah)		**4.30**				

Compare against your selected battery (e.g. 2x 12V 7Ah = 7Ah per string). This recalculates automatically as you tune load assumptions above.

Open Decisions / Assumptions Log

#	Item	Current assumption in this workbook	Status
1	SMPS module output voltage/current	30V DC raw (per your confirmation, headroom for 28.2V float charging). Current rating is CALCULATED in Power_Tree, not assumed — see 'Recommended SMPS rating' near the bottom of that sheet.	RESOLVED
2	RS-485 transceiver	MAX485ESA+ (5V) kept, with a 3.3V<->5V level shifter added between it and STM32G0B1's UART.	RESOLVED
3	Shift registers / second MCU	Removed. Single STM32G0B1CBT6 drives HMI directly via GPIO — see Architecture sheet GPIO budget.	RESOLVED
4	Field detector current draw (25-30 devices)	RESOLVED with real spec: 2-wire/24VDC, standby <=0.1mA, alarm 25mA. Zone loop current is now a live formula (EOL resistor + device count) in Power_Tree, not a copied benchmark.	RESOLVED
5	External LCD module exact part	RESOLVED: JHD162A, +5V, 1.5mA per datasheet.	RESOLVED
6	LCD interface type	LCD supporting I2C pins are used. No external backpack is needed.	**OPEN — decide: add a PCF8574 I2C backpack to the LCD (frees 4 pins), or free pins elsewhere (fewer LEDs / matrix the keypad)**
7	RL1/RL2 — NAC1/NAC2 relays	RESOLVED: confirmed present in the reverse-engineered netlist (RL1 -> NAC1, RL2 -> NAC2), same HFD27/024 spec as Fire/Fault/Supervisory. Added to Quantity_BOM (relay qty 3->5) and Power_Tree (24V rail).	RESOLVED

8	RS-SEN and RS-24VOP	RESOLVED from the actual .kicad_sch: both are plain 2-pin terminal blocks (J12, J13), not relay outputs. RS-24VOP = direct 24V output tap (likely repeater panel power). RS-SEN = supervised sense input (R69 820k/R70 4.7k divider into the analog mux U12/U13), monitors a remote link's presence/health. Added to Quantity_BOM and Power_Tree.	RESOLVED
9	Battery capacity vs updated load model	Power_Tree recomputes required Ah live from the current load model. Relay coil voltage (item 11) is now resolved at 24V, so this figure is stable — recheck again only if detector count/spec or LCD assumptions change further.	RESOLVED — recheck if load assumptions change again
10	LM317T charger headroom at 30V input	27.6-28.2V float target from 30V raw leaves only ~1.8-2.4V headroom. LM317 typical dropout is ~2-3V at rated current — fine at the charger's modest 0.8A max, but confirm against the specific LM317T grade/heatsinking you use.	**OPEN — verify at bring-up**
11	Relay coil voltage — K1-K5 (G5V-2)	RESOLVED: 24V coil, per your decision. Rationale: fail-safe relays (Fault/Supervisory) should draw power from the rail closest to the battery, not through 2 extra regulation stages; less wasted heat at 24V than 5V for the same coil power; keeps the already-busier 5V rail lighter. Order the 24VDC-coil variant of G5V-2.	RESOLVED

12	TIP127 vs BDX34C relay driver	RESOLVED in favour of TIP127, per your direct visual inspection of the physical Ravel board — this outranks the other engineer's .kicad_sch, which showed BDX34C at this position (likely a transcription slip or a substitution on their end, not confirmed). Checked both datasheets: same TO-220 package, same pinout, same 100V rating, both far exceed what a relay coil actually draws — so even if BDX34C is what's really there, no functional consequence. Quantity_BOM reverted to TIP127.	RESOLVED
13	L7805CV output tolerance vs MAX485ESA+ Vcc spec	L7805CV nominal 5V, typ. +/-4% class (generic L78xx family) puts the rail at roughly 4.8-5.2V. MAX485ESA+ datasheet Vcc spec is 4.75-5.25V. The rail's 4.8V floor clears the transceiver's 4.75V floor by only 0.05V on this generic tolerance figure — the tightest margin anywhere in the power tree. This workbook has not been given the specific L7805CV manufacturer datasheet to confirm the real tolerance grade.	**OPEN — confirm exact L7805CV grade/tolerance against its datasheet before BOM lock; consider a tighter-tolerance 5V regulator if margin is a concern**
14	STM32G0B1 UART pins are 5V-tolerant (FT) — possible level-shifter simplification	STM32G0B1CBT6 datasheet Table 12 marks every I/O used for USART/RS-485 (PC10/PC11 etc.) as FT (5V tolerant), including under alternate function. This means the MCU's RX input can likely accept the MAX485ESA+ 5V logic output directly, without needing the 3.3V<->5V level shifter on that leg. The MCU's TX output (3.3V high, ~3.0V min per FT pin spec) driving into MAX485ESA+ is the side that still needs checking against MAX485's guaranteed VIH.	**OPEN — your call: keep the bidirectional level shifter as designed (safest, no change needed), or investigate a single-direction (TX-only) level shift to simplify U3. Not changed in this workbook pending your decision.**